LLM-powered-meme-responder: v1 vs v2 Summary (repo evidence only)

Date: 2026-02-09 | Scope: README.md, v1/, v2/, tags.jsonl

What it is

LLM-powered-meme-responder is a Python project that turns a free text prompt into a meme style response. The repo contains a runnable v1 pipeline and a modular v2 redesign focused on explainable decision stages.

v1 - What it does

- Normalizes the prompt, then loads meme metadata and image assets from local files (v1/src/pipeline/generate.py, v1/memes/metadata.jsonl).
- Selects a meme by trigger phrase match first, then falls back to embedding similarity over v1/indexes/meme_emb.npy.
- Calls a local Ollama endpoint (127.0.0.1:11434) with qwen2.5:3b-instruct to generate a short JSON caption.
- Renders bottom caption text onto the selected image using Pillow and writes v1/outputs/out.png.
- Includes a meme indexing script that encodes tag plus trigger text with all-MiniLM-L6-v2.

v1 - How it works (repo evidence)

Flow: prompt -> normalize -> trigger match OR embedding retrieval -> meme id -> LLM caption -> image render. Core files: v1/src/pipeline/generate.py, v1/src/llm/client.py, v1/src/render.py, v1/scripts/build_meme_index.py.

v1 - Not found in repo

- CLI and Gradio entrypoints are present but empty (v1/src/ui/cli.py, v1/src/ui/gradio_app.py).
- Quote subsystem data and scripts are present but empty (v1/quotes/quotes.jsonl, v1/scripts/build_quote_index.py, quote index files).
- Test files exist but contain no assertions (v1/tests/test_*.py).

v2 - What it does

- Defines four modules: ModuleA classifier, ModuleB policy engine, ModuleC generator, ModuleD retrieval (README.md, v2/Module*/).
- ModuleA loads tags from tags.jsonl, normalizes prompts, trains grouped TF-IDF plus logistic models, and predicts tags.
- ModuleB loads and validates policy rules, scores by overlap times weight, and returns a ReactionPlan.
- ModuleC loads templates, renders slot values, and applies constraints (length cap, no mentions or hashtags, emoji removal).
- A JSON schema exists for policy rules in v2/ModuleB/policy_rules.schema.json.
- README marks v2 as architecture first; implementation still in progress.

v2 - How it works (repo evidence)

Flow: prompt -> ModuleA tags -> ModuleB reaction plan -> ModuleC response text -> ModuleD meme template ranking and selection. Current status: ModuleA and much of ModuleB are implemented; ModuleC generate_response and most ModuleD functions raise NotImplementedError.

v2 - Not found in repo

- No end to end runner or API entrypoint under v2/.
- No sample v2 policy rule file, response template file, or meme catalog file in v2/.

v1 vs v2 - Differences

- Maturity: v1 executes an end to end path; v2 is partially implemented.
- Design: v1 is mostly monolithic; v2 separates classification, policy, generation, and retrieval contracts.
- Decisioning: v1 uses trigger and embedding retrieval directly; v2 adds explicit tag and rule based reaction planning.
- Explainability: v2 keeps interpretable intermediates (tags and reaction plan); v1 mainly outputs meme plus caption.
- Gaps: v1 lacks implemented UI, quotes, and tests; v2 lacks completed generation, retrieval, and orchestration.